

Instructor Key: Day 6 Mastery Quiz: Branch Routing

Branch Routing

Key for the Week 2 student quiz package. Keep instructor-facing.

1 Artifact status and source boundary

Field	Status
Student artifact	10-point Day 6 mastery quiz; quiz, not workbooklet.
Content anchor	Built from the locally verified Week 2 workbooklet content anchor for sensor-kit checkout and release routing.
Difficulty posture	Shorter and sharper than the workbooklet. It asks for trace, route, debug, test, and transfer evidence.
Student-surface rule	Student PDF hides internal file versioning and release/validation language; visible student surface carries Lance copyright.
Canvas status	Not Canvas-released. Student PDF still needs Lance upload/staging plus Ally and Panorama review if selected for use.

2 Reference code

```
1 kit_label = "S-21"
2 charge_percent = 76
3 charge_minimum = 80
4 calibration_label = "current"
5 borrower_initials = "RG"
6
7 charge_ok = charge_percent >= charge_minimum
8
9 if charge_ok:
10     action = "send to calibration check"
11 else:
12     action = "charge before checkout"
13
14 route_note = kit_label + ": " + action
```

3 Point map, secure answers, and support moves

Pt.	Item target	Secure answer	Support move
1	1. charge_ok	False. The comparison is $76 \geq 80$, which is false.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	2. branch followed	The else branch runs because charge_ok is false.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	3. action	"charge before checkout".	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	4. route_note	"S-21: charge before checkout".	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	5. boundary case	True. The comparison uses $\geq$, so $80 \geq 80$ passes.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	6. changed charge	The if branch runs and action becomes "send to calibration check".	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	7. scope discipline	calibration_label is stored, but no condition in this code uses it to choose the branch.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	8. branch rule	Only one branch runs: if the condition is true the if block runs; otherwise the else block runs.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	9. evidence sentence	Secure answer names false condition, else branch, and charge-before-checkout action.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	10. support route	Return to condition-result-first tracing; write True/False before naming the branch.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.

4 Mastery interpretation

Band	Point guide	Evidence pattern	Next support action
Secure	8–10 / 10	Student names the route evidence, controlling condition, and transfer/debug implication.	Move to the next daily packet or Exam 1/Week 3 bridge as appropriate.
Developing	5–7 / 10	Student can get final route but may not preserve condition order or expected/actual evidence.	Use a route ledger: condition, value, branch, action, reason.
Needs support	0–4 / 10	Student guesses route from scenario language or final output without tracing.	Return to first condition, true/false result, branch selected, and support note.

5 Release/accessibility notes

Local verification should check tagged PDF output, extractable text, page count, render sheets, answer-key separation, no private/protected markers, and the student-facing no-internal-surface rule. If selected for Canvas use, Lance uploads/stages the student PDF and retrieves both Ally and Panorama scores.